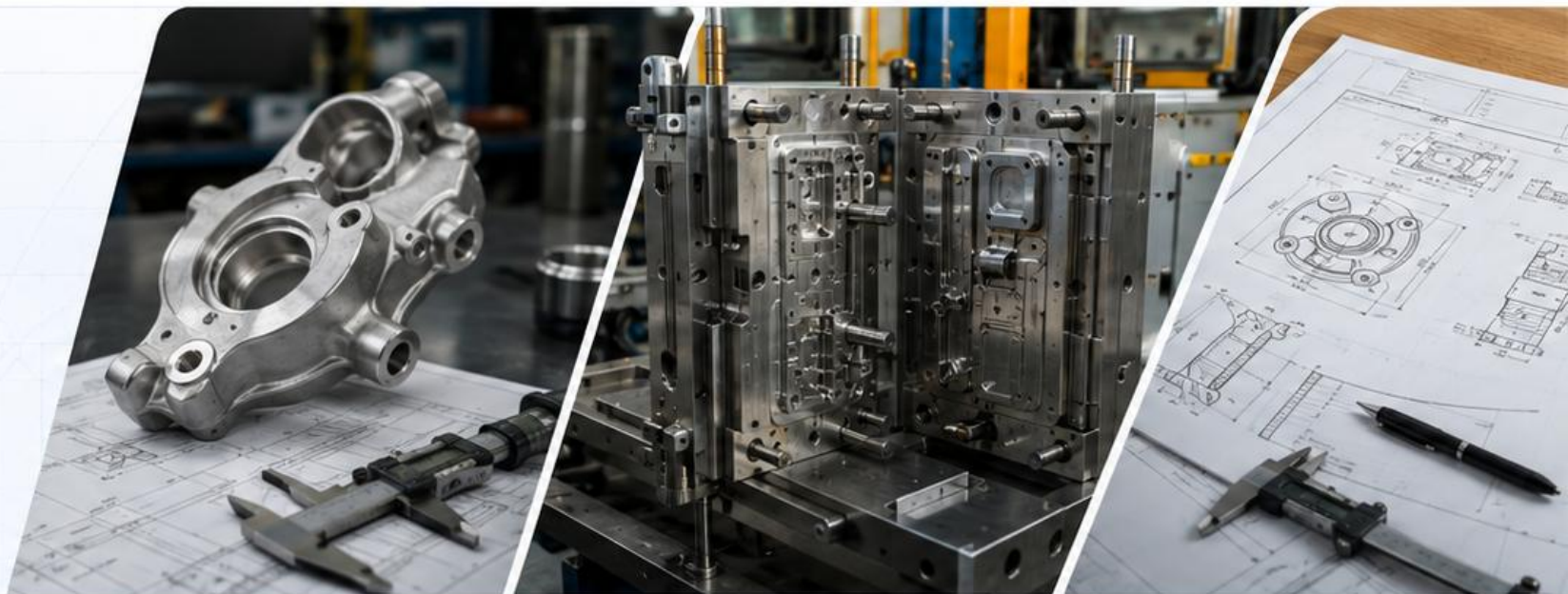


# ENGINEERING DESIGN BUILT FOR MANUFACTURING

Practical mechanical design support for products, tooling, CAD documentation, and supplier coordination.

 Windsor, Ontario, Canada

 15+ Years of Engineering Experience



## Product & Mechanical Design

3D CAD Modelling

Drawings, GD&T & BOMs



## Mold & Die Tooling Design

3D CAD Modelling

Drawings, GD&T & BOMs



## CAD Documentation & Manufacturing Support

DFM/DFA

Supplier Coordination



STEP 01

### Understand

Clarify requirements and functional needs.



STEP 02

### Develop

Explore concepts and evaluate solutions.



STEP 03

### Detail

Create detailed models and engineering documentation.



STEP 04

### Review

Validate designs and manufacturability.



STEP 05

### Support

Support production and answer supplier questions.



## LET'S REVIEW YOUR ENGINEERING PROJECT

Discuss your design requirements, documentation needs, or supplier coordination challenges.

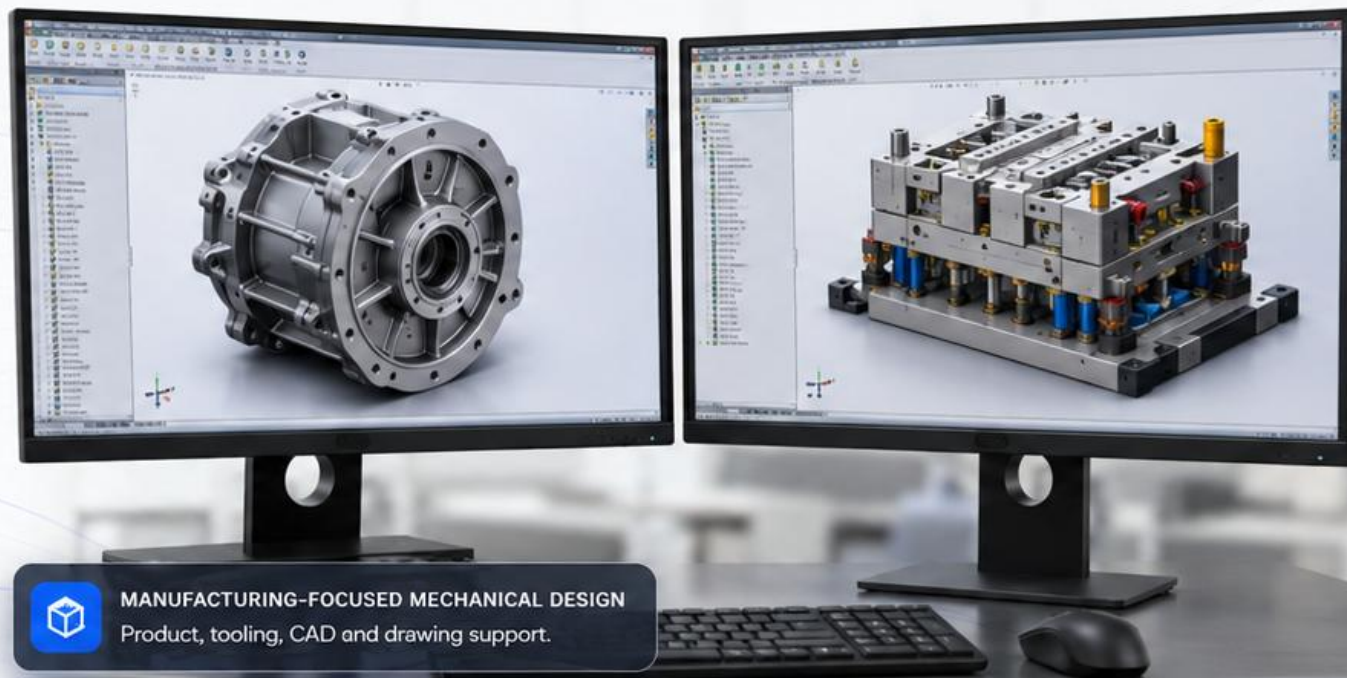
[Request a Project Review →](#)
[Explore Services →](#)



## ABOUT AG VERTEX

# ENGINEERING DESIGN BUILT FOR MANUFACTURING

AG Vertex is a Windsor, Ontario-based mechanical design consultancy specializing in product design, mold and die design, 3D CAD modelling, and automotive drawing validation. We help transform engineering concepts into practical, manufacturable designs supported by accurate documentation and supplier coordination.

[Explore Our Capabilities →](#)

### MANUFACTURING-FOCUSED MECHANICAL DESIGN

Product, tooling, CAD and drawing support.



### OUR MISSION

To deliver practical and precise mechanical design solutions that support manufacturability, quality, and efficient product development.



### OUR VISION

To become a trusted engineering design partner for manufacturers, tooling companies, and automotive suppliers across Canada and beyond.



### CORE VALUES

Technical Integrity · Quality ·  
Collaboration · Confidentiality ·  
Continuous Improvement



### OUR PROMISE

Clear communication, carefully developed deliverables, and engineering support focused on each client's technical and manufacturing requirements.



# 15+ YEARS OF PROFESSIONAL ENGINEERING EXPERIENCE

AG Vertex is built on experience across product development, injection molding, hot-runner systems, automotive components, and manufacturing-focused mechanical design.



## PRODUCT DEVELOPMENT

Mechanical components and assemblies developed with performance, manufacturability, and production requirements in mind.



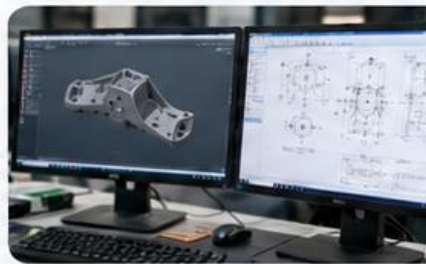
## TOOLING EXPERIENCE

Practical experience supporting injection molds, hot-runner systems, mold components, and pressure die-casting applications.



## AUTOMOTIVE ENGINEERING

Experience with suspension, steering, wheel-end components, engineering drawings, GD&T, and supplier coordination.



## CAD & DOCUMENTATION

3D models, drawings, BOMs, and controlled documentation using established CAD and PLM workflows.



## 15+ YEARS

Professional Engineering Experience



## 3D CAD

Parts, Assemblies & Tooling



## GD&T / DFM

Manufacturing-Focused Design



## SUPPLIER SUPPORT

Drawing Review & Coordination

START A CONVERSATION

## DISCUSS YOUR ENGINEERING PROJECT

Let's review your requirements and define a practical design path forward.

[Request a Project Review →](#)



### AG VERTEX

Mechanical design consultancy in Windsor, Ontario.



### SERVICES

Product Design & Development  
Injection Mold Design  
Pressure Die-Casting Die Design  
3D CAD Modelling  
Drawings, GD&T & BOMs  
Automotive Drawing Validation

### COMPANY

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### CONTACT

[Windsor, Ontario, Canada](#)  
[contact@agvertex.com](mailto:contact@agvertex.com)



## OUR CAPABILITIES

# ENGINEERING DESIGN SERVICES

Manufacturing-focused mechanical design support—from product development and tooling design to detailed CAD documentation and supplier coordination.



## 01

### DEVELOP



#### PRODUCT DESIGN & DEVELOPMENT

Concept development, component design, DFM/DFA and production-focused engineering support.

[View Details →](#)



#### 3D CAD MODELLING

Detailed parts and assemblies, CAD conversion, design modifications and parametric modelling.

[View Details →](#)



## 03

### DOCUMENT



#### DRAWINGS, GD&T & BOMs

Manufacturing drawings, tolerance definition, GD&T application and structured bills of materials.

[View Details →](#)



#### AUTOMOTIVE DRAWING REVIEW

Review of dimensions, GD&T, specifications, interfaces and supplier drawing questions.

[View Details →](#)



## 02

### TOOLING



#### INJECTION MOLD DESIGN

Mold layouts, core and cavity development, inserts, slides, lifters and manufacturing documentation.

[View Details →](#)



#### PRESSURE DIE-CASTING DIE DESIGN

Die concepts, parting-line strategy, core and slide concepts, and detailed die documentation.

[View Details →](#)



## 04

### COORDINATE



#### DFM / DFA SUPPORT

Design reviews focused on manufacturability, assembly, cost awareness and production feasibility.

[View Details →](#)



#### SUPPLIER & PROTOTYPE COORDINATION

Technical coordination, DFM clarification, prototype follow-up and engineering change support.

[View Details →](#)

## 1

#### REQUIREMENTS

Understand goals, constraints and technical requirements.



## 2

#### CAD DEVELOPMENT

Create models, assemblies and detailed design solutions.



## 3

#### DRAWING REVIEW

Validate design intent, GD&T and documentation accuracy.



## 4

#### SUPPLIER COORDINATION

Support manufacturing handoff, prototypes and engineering changes.

## HAVE A PROJECT TO DISCUSS?

Let's review your design requirements, documentation needs, or supplier drawing challenges.

[Request a Project Review →](#)

## INDUSTRIES WE SUPPORT

# ENGINEERING SUPPORT FOR MANUFACTURING

Practical mechanical design, tooling, drawing, and supplier support across automotive and manufacturing sectors.

[Automotive Components](#)
[Injection-Molded Products](#)
[Industrial Products](#)
[Electromechanical Components](#)

## AUTOMOTIVE COMPONENT DESIGN SUPPORT

From concept development to production-ready engineering documentation.

3D component design and detailed documentation for suspension, steering, wheel-end, molded-plastic, and electromechanical parts—developed with manufacturability and supplier coordination in mind.



Product Development



GD&T / DFM



Supplier Coordination



AUTOMOTIVE COMPONENT DESIGN & DOCUMENTATION

### ⚠️ TYPICAL CLIENT NEEDS

- Production-ready component design
- Drawing discrepancies and design changes
- Manufacturing feasibility and supplier alignment

### ✅ HOW AG VERTEX SUPPORTS

- ✅ 3D CAD modelling and design refinement
- ✅ Drawing review, GD&T, and documentation
- ✅ Supplier coordination and issue resolution

[REQUEST AN AUTOMOTIVE PROJECT REVIEW →](#)


Mechanical design consultancy in Windsor, Ontario.



### SERVICES

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Drawings, GD&T & BOMs

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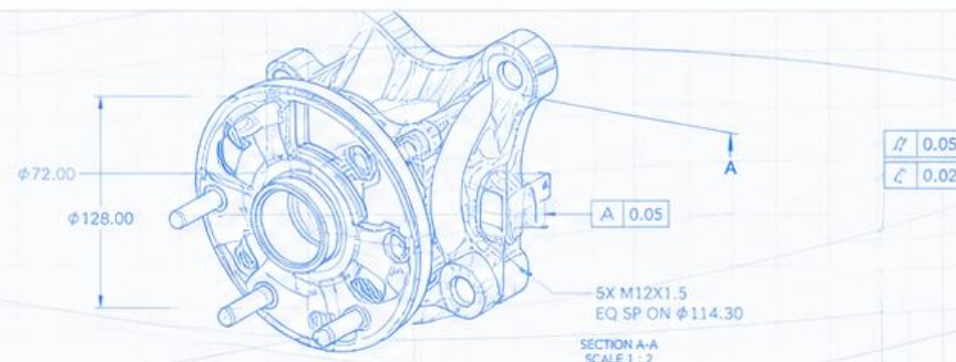
📍 Windsor, Ontario, Canada  
✉️ [contact@agvertex.com](mailto:contact@agvertex.com)



## ENGINEERING CAPABILITY SHOWCASE

# MANUFACTURING-FOCUSED DESIGN CAPABILITIES

Representative examples of product design, tooling, CAD modelling, drawing review, and supplier support.



### CAPABILITY INDEX



All  
Capabilities



Product Design



Mold & Die Design



3D CAD



Drawing Review



#### PRODUCT DESIGN

### Automotive Component Development

Concept development and detailed design of automotive components.



#### DRAWING REVIEW

### Automotive Drawing Review

Drawing verification, discrepancy resolution, and supplier coordination.



#### MOLD & DIE DESIGN

### High-Pressure Die-Casting Tool Design

High-pressure die-casting tool concepts, parting lines, and slide mechanisms.



#### MOLD & DIE DESIGN

### Injection Mold Tooling

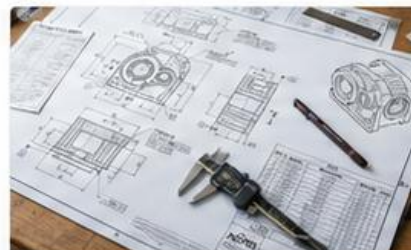
Mold design, component detailing, and tooling support for injection molding.



#### 3D CAD

### Parametric CAD & Assembly Design

Parametric modeling, assembly design, and CAD data preparation.



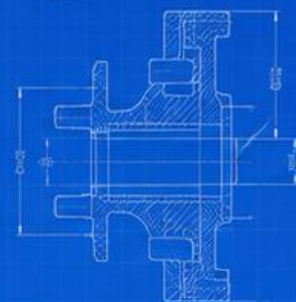
#### DRAWING REVIEW

### ASME Y14.5 GD&T & BOM Drafting

Production-ready drawings with GD&T per ASME Y14.5 and structured bills of materials.

**Need support on a design or tooling project?**

Our engineering team can help from concept through production support.



[Request a Project Review →](#)

Representative capability visuals. Client work is shown only with authorization.



## ENGINEERING KNOWLEDGE HUB

# Resources & Insights

Practical guidance on product design, tooling, 3D CAD, GD&T, DFM and automotive drawing review.

[All](#)
[Product Design](#)
[Mold Design](#)
[CAD & Drawings](#)
[Automotive](#)

### FEATURED PRACTICAL GUIDE

## Automotive Drawing Review: A Practical Checklist

Key checks for dimensions, GD&T, materials, revisions, manufacturability and supplier clarification.

⌚ 6 min read | 🚗 Automotive

[Read Guide →](#)


### Designing Plastic Parts for Injection Molding

Practical DFM checks for wall thickness, draft, ribs, bosses and parting line strategy.

[Read Article →](#)


### GD&T Drawing Review: Common Issues to Check

A focused review of datums, tolerances, feature control frames and drawing clarity.

[Read Article →](#)


### From 3D CAD Model to Production Drawing

How modelling decisions, tolerances, BOMs and revision control support manufacturing.

[Read Article →](#)


### Need guidance for your project?

Discuss your requirements with AG Vertex.

[Request a Project Review →](#)



## CAREERS & COLLABORATION

# BRING PRACTICAL ENGINEERING IDEAS TO LIFE

AG Vertex welcomes experienced mechanical designers, tooling specialists, and CAD professionals interested in future project-based collaboration.

[Submit Your Profile →](#)


### PRACTICAL ENGINEERING

We solve real design problems with practical, manufacturable solutions.



### FLEXIBLE COLLABORATION

Work with us on a project basis or as an independent specialist.



### TECHNICAL INTEGRITY

We stand for accuracy, reliability, and clear communication in every deliverable.



### CONTINUOUS LEARNING

We encourage knowledge sharing and ongoing growth in engineering.



## NO ACTIVE POSITIONS AT THIS TIME

We are not currently hiring for permanent roles. You may submit your profile for consideration when suitable project-based opportunities arise.

## AREAS OF INTEREST



Product Design & Development



Mold & Die Design



3D CAD Modelling



Drawings, GD&T & BOMs



Automotive Drawing Review

## INTERESTED IN FUTURE PROJECT-BASED WORK?

Submit your profile and we'll keep you in mind for suitable project-based opportunities.

[Submit Your Profile →](#)


Mechanical design consultancy in Windsor, Ontario, Canada.



### SERVICES

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### CONTACT

Windsor, Ontario, Canada  
info@agvertex.com

[Request a Project Review →](#)



GET IN TOUCH

# Let's discuss your project

Tell us what you are developing, the engineering support you need, and your preferred timeline.

## Request a Project Review

Name

e.g. Alex Mercer

Work Email

e.g. alex@company.com

Phone Number (Optional)

e.g. (519) 555-0123

Service Required

Select a service

Project Overview

Briefly describe your component, tooling, drawing, or CAD requirements...

Preferred Timeline (Optional)

Select a timeline

☐ I agree that AG Vertex may contact me about this inquiry.

Send Project Request

## AG Vertex

Windsor, Ontario, Canada

info@agvertex.com

Project inquiries by appointment

### Engineering Support

Product Design · Mold & Die Design · 3D CAD Modelling · Drawings, GD&T & BOMs · Automotive Drawing Review

### What happens next

- 1 We review your project brief
- 2 We clarify scope and timing
- 3 You receive the recommended next step



### Services

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